

Case Study Group A

Does Annual Household Income Play a Role in Minority's Political Party Identification?

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Research question

The research question is: **"Does Annual Household Income Play a Role in Minority's Political Party Identification?"**

- 1. Try to explain why this research question might be of interest and to whom.**

This research question is relevant because it challenges a common assumption: that income is a primary driver of political preferences.

This question that has implications for several stakeholders:

- a. Political Parties and Campaign Strategists:** Knowing whether income drives minority voters' party identification would help parties design more effective outreach strategies and determine whether economic messaging or culturally relevant approaches are more appropriate.
- b. Policy Makers:** Understanding the relationship between income and political preferences among minorities can help shape more inclusive and representative policy agendas.
- c. Academic Researchers:** This question contributes to the broader debate in political science about the determinants of voting behavior. While traditional models emphasize economic factors such as income and social class, this study explores whether cultural and ethnic identity may play a more significant role among minority populations in the UK.
- d. Ethnic Minority Communities:** Exploring what shapes their political identity helps reinforce the importance of cultural representation in politics.
- e. Media and Public Opinion Analysts:** Understanding the real factors behind minority voting patterns helps avoid oversimplified narratives that reduce political behavior to income levels.

- 2. Locate relevant data file and accompanying documentation in the "stata" folder.**

- Data file: BSE_2010.dta
Format: Stata (.dta)
Source: British Election Study - Ethnic Minority Survey 2010
- Documentation:
british_election_study_ethnic_minority_survey_2010_data_anon_ukda_data_dictionary.rtf
This data dictionary contains the description of each variable, including variable names, labels, and the meaning of each coded value.
- Technical Report:
british_election_study_ethnic_minority_survey_2010_technical_report.pdf

f. Published by TNS-BMRB (January 2012). Contains the full methodology of the study.

3. Perform any necessary transformations of data format type.

The following transformations were performed on the dataset:

- a. **Reading the .dta file:** The dataset was in Stata format (.dta). It was read using pandas with `convert_categoricals=False` to avoid errors caused by duplicate value labels in the file.

```
df_original = pd.read_stata('BSE_2010.dta', convert_categoricals=False)
```

- b. **Column renaming:** The original column names were coded (e.g., `zq88`, `bq9_1`, `zqinc`). A copy of the DataFrame was created and columns were renamed using the variable labels stored in the .dta file to make them readable (e.g., `zq88` -> Gender, `zqinc` -> Annual Household Income).

```
df_labeled = df_original.copy() df_labeled.rename(columns=labels, inplace=True)
```

- c. **Value label mapping:** The data contained numeric codes representing categories (e.g., 1, 2, 3 instead of Indian, Pakistani, Bangladeshi). New columns were created mapping these codes to their text labels for readability in crosstabs and visualizations. Examples:

-Five Ethnicities: 1=Indian, 2=Pakistani, 3=Bangladeshi, 4= Black Caribbean, 5=Black African

- Annual Household Income: 1=£0-5,000, 2=£5,001-10,000, etc.

- d. **Filtering invalid values:** Categories -2 (Refused) and -1 (Don't Know) were filtered out from relevant variables to ensure only valid responses were included in the Chi-Square analysis. The numeric column was used for filtering (e.g., `df_labeled ['Annual Household Income'] > 0`) while the label columns were used for visualization.

4. Identify variables of interest and specify their type.

	Variable	Type
0	Unique ID	Quantitative
1	Country	Qualitative
2	Gender	Qualitative
3	Age	Quantitative
4	Party Identification	Qualitative
5	Closer to Party	Qualitative
6	Importance of Religion	Qualitative
7	Ethnicity	Qualitative
8	Feelings-Labour Party	Quantitative
9	Feelings-Conservative Party	Quantitative
10	Feelings-Liberal Democrats	Quantitative
11	Feelings-Green Party	Quantitative
12	Feelings_BNP	Quantitative
13	Feelings-UKIP	Quantitative
14	Feelings-SNP	Quantitative
15	Feelings-Plaid Cymru	Quantitative
16	Democracy Satisfaction	Qualitative
17	Pol Activity-Good Way Benefits Self & Family	Qualitative
18	Pol Activity-Too Much Time & Effort	Qualitative
19	Pol Activity-Good Way Group Benefits	Qualitative
20	Annual Household Income	Qualitative
21	Party Vote 2010	Qualitative
22	Party Best For Ethnic Minorities	Qualitative
23	Five Ethnicities	Qualitative

5. In case a variable is ordinal or nominal, list the possible categories and their meaning.

	Variable	Code	Subtype	Categories
1	Country	Country	Nominal	1=ENGLAND; 2=SCOTLAND; 3=WALES
2	Gender	zq88	Nominal	1=Male; 2=Female; 3=Not Stated
4	Party Identification	bq9_1	Nominal	-2=Refused; -1=Don't Know; 1=None/No; 2=Labour...
5	Closer to Party	bq9_2	Nominal	-2=Refused; -1=Don't Know; 1=Yes; 2=No
6	Importance of Religion	eq4	Ordinal	-2=Refused; -1=Don't Know; 1=Extremely importa...
7	Ethnicity	zq101	Nominal	-2=Refused; -1=Don't Know; 1=White - British; ...
16	Democracy Satisfaction	bq61	Ordinal	-2=Refused; -1=Don't Know; 1=Very satisfied; 2...
17	Pol Activity-Good Way Benefits Self & Family	bq18_3	Ordinal	-2=Refused; -1=Don't Know; 1=Strongly agree; 2...
18	Pol Activity-Too Much Time & Effort	bq18_4	Ordinal	-2=Refused; -1=Don't Know; 1=Strongly agree; 2...
19	Pol Activity-Good Way Group Benefits	bq18_5	Ordinal	-2=Refused; -1=Don't Know; 1=Strongly agree; 2...
20	Annual Household Income	zqinc	Ordinal	-2=Refused; -1=Don't Know; 1=0 - 5000; 2=5001 ...
21	Party Vote 2010	bq12_2	Nominal	-2=Refused; -1=Don't Know; 1=Labour; 2=Conserv...
22	Party Best For Ethnic Minorities	bq71	Nominal	-2=Refused; -1=Don't Know; 1=Labour; 2=Conserv...
23	Five Ethnicities	ScreenEthQairre	Nominal	1=Indian; 2=Pakistani; 3=Bangladeshi; 4=Black ...

6. Define the analysis required to answer the RQ and explain why you have chosen this particular analysis.

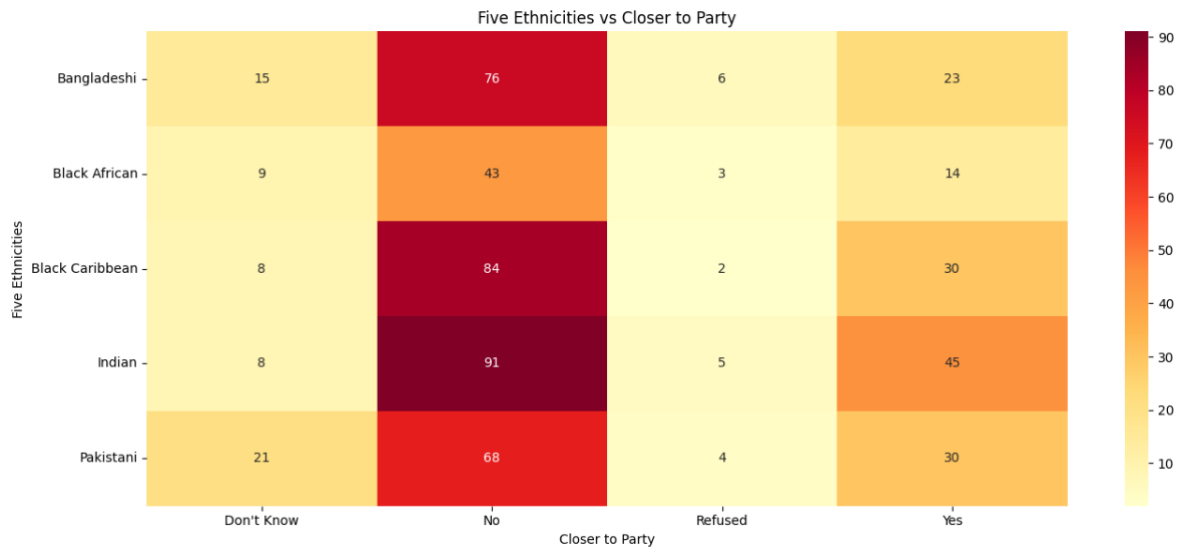
The analysis chosen is the Chi-Square Test of Independence. This test evaluates whether two categorical variables are independent or if there is a statistically significant association between them.

This test was selected because:

- All variables involved are categorical
- The research question asks whether income "influences" party identification, which requires testing if these variables are independent or associated.
- Chi-Square is the standard statistical test for evaluating independence between two categorical variables using a crosstab.

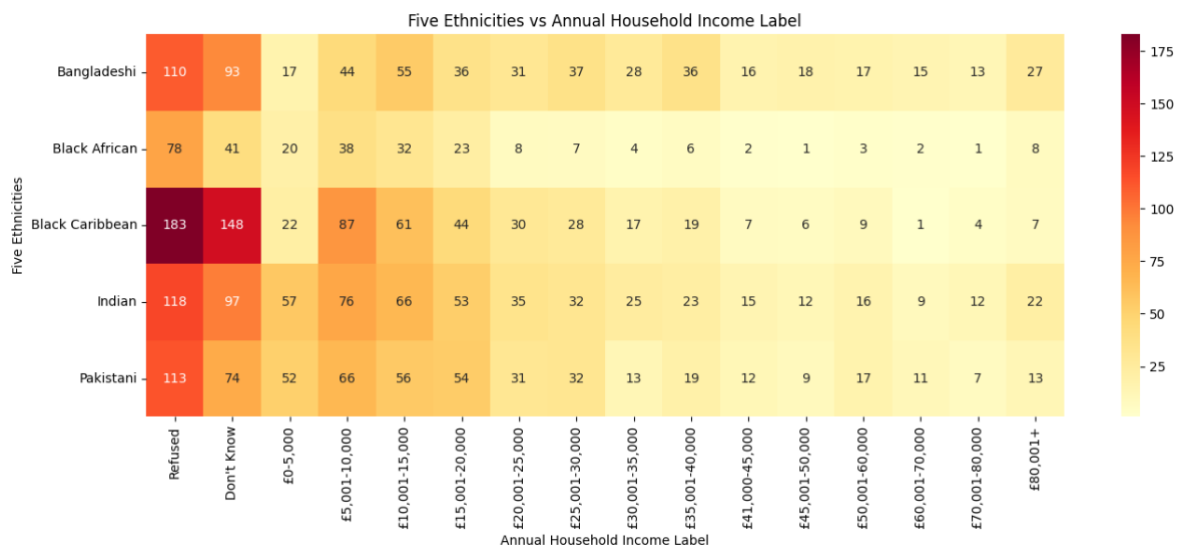
The following pairs of variables were tested:

- Ethnicity vs Closer to Party:** to explore if ethnicity influences whether a person feels close to any party.
 - chi_square statistic: 19.450
 - p_value:0.07824
 - dof:12



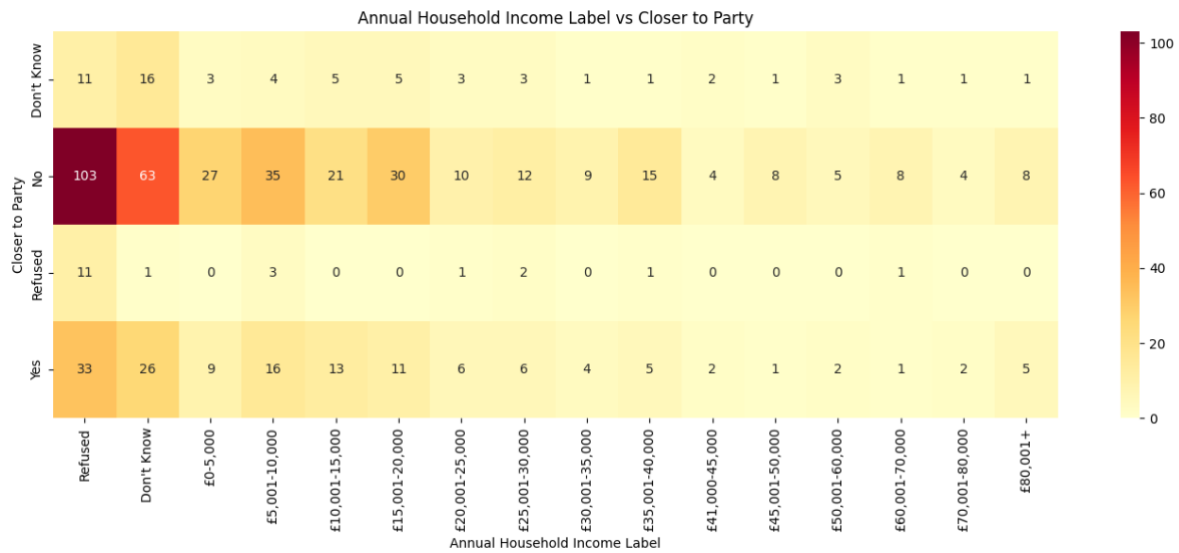
b. Income vs Ethnicity: to determine if income distribution varies across ethnic groups.

- chi_square statistic: 194.163
- p_value:0.00000
- dof:60



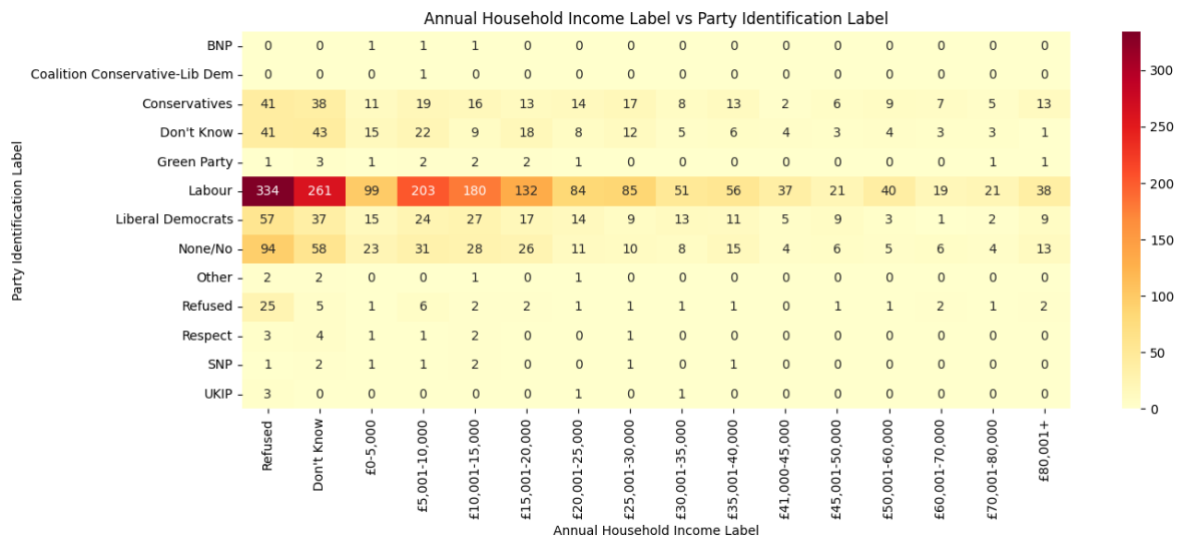
c. Income vs Closer to Party: to determine if income influences whether a person feels close to any political party.

- chi_square statistic: 38.955
- p_value:0.72478
- dof:45



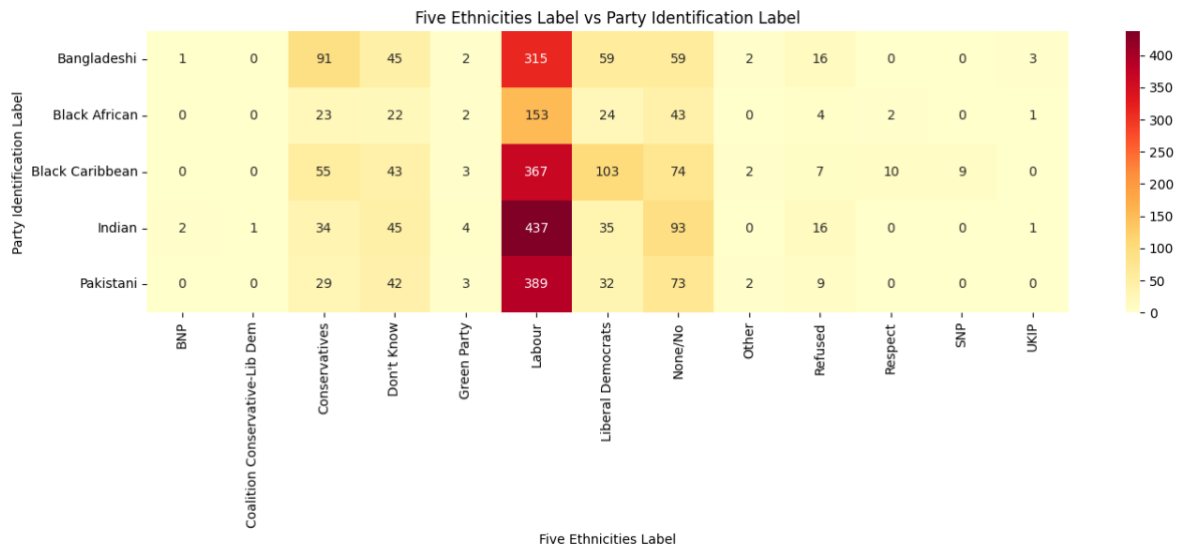
d. Income vs Party Identification: to directly answer the RQ whether income influences which party minorities identify with.

- chi_square statistic: 182.160
- p_value: 0.44097
- dof: 180



e. Ethnicity vs Party Identification: to explore if ethnicity plays a role in party identification.

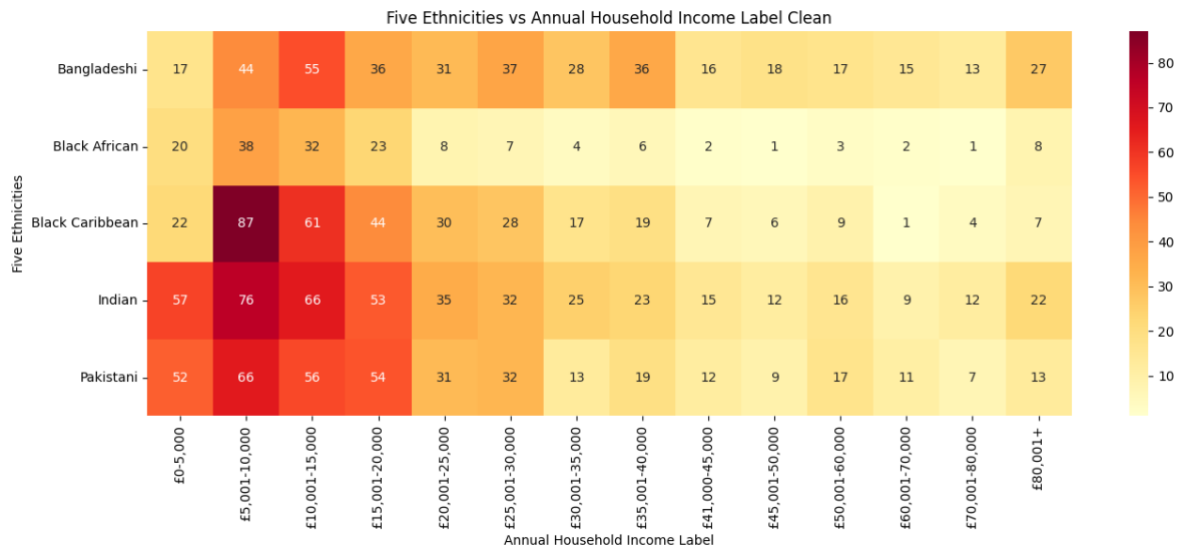
- chi_square statistic: 203.141
- p_value: 0.00000
- dof: 48



Additionally, each income-related test was run twice: once including all responses and once excluding -2 (Refused) and -1 (Don't Know) values, to verify that invalid categories were not inflating the results.

f. Income Clean Vs Ethnicity:

- chi_square statistic: 130.093
- p_value:0.00000
- dof:52



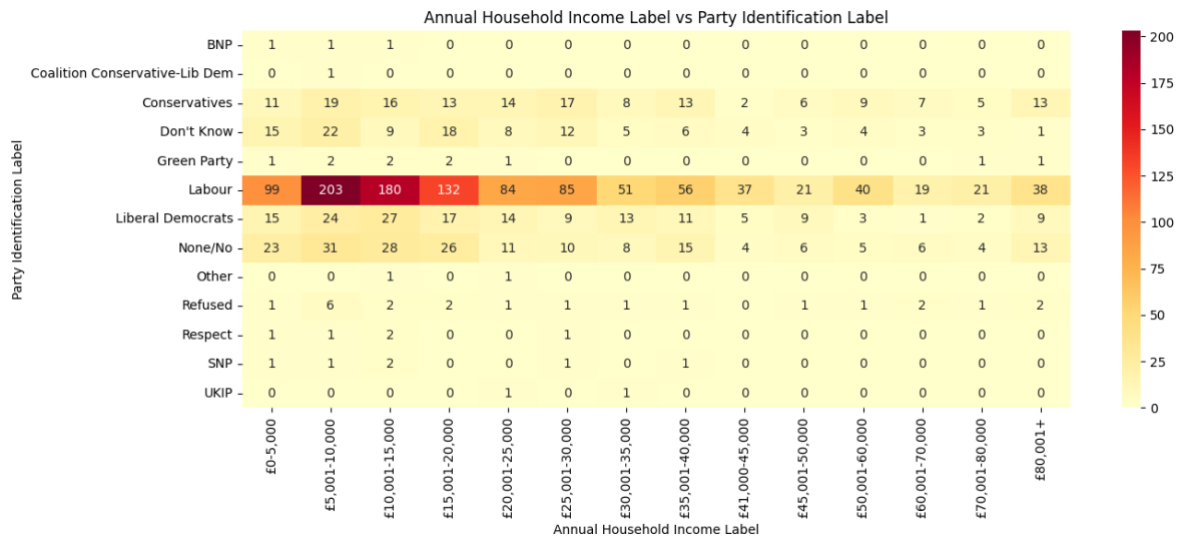
g. Income Clean vs Closer to Party

- chi_square statistic: 27.983
- p_value:0.90518
- dof:39



h. Income Clean vs Party Identification

- chi_square statistic: 134.265
- p_value:0.89518
- dof:156

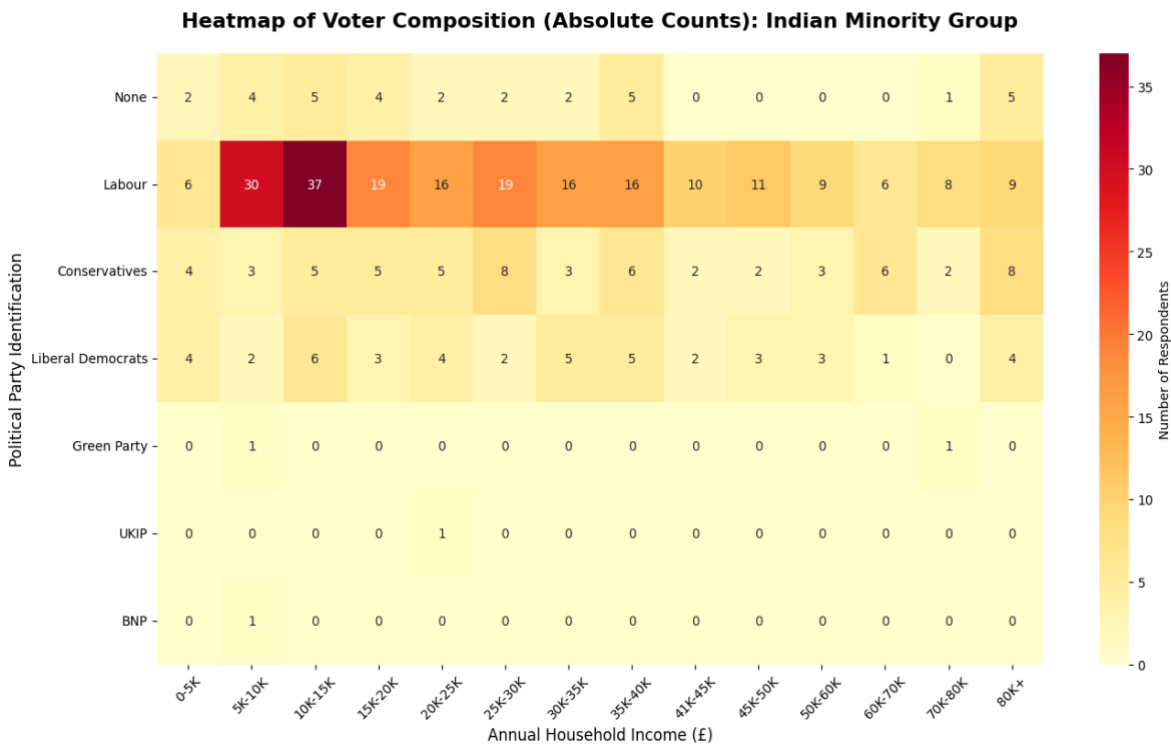


In-depth Analysis: specific ethnic group breakdown

While the aggregate analysis of the total population provides a statistically robust baseline, it risks masking significant internal variations through the "averaging out" of diverse cultural experiences. To ensure the integrity of our conclusions, it is essential to conduct a disaggregated analysis of individual ethnic minority groups.

The purpose of this in-depth breakdown is twofold: to verify if the "income-neutral" voting pattern is a universal characteristic among minorities or if specific groups deviate from the aggregate trend, and to provide a granular view of the relationship between socioeconomic mobility and political loyalty, thereby testing the limits of the class-alignment theory across different cultural backgrounds.

- **Indian Population:**
- chi_square statistic: 80.379
- p_value:0.40444
- dof:78



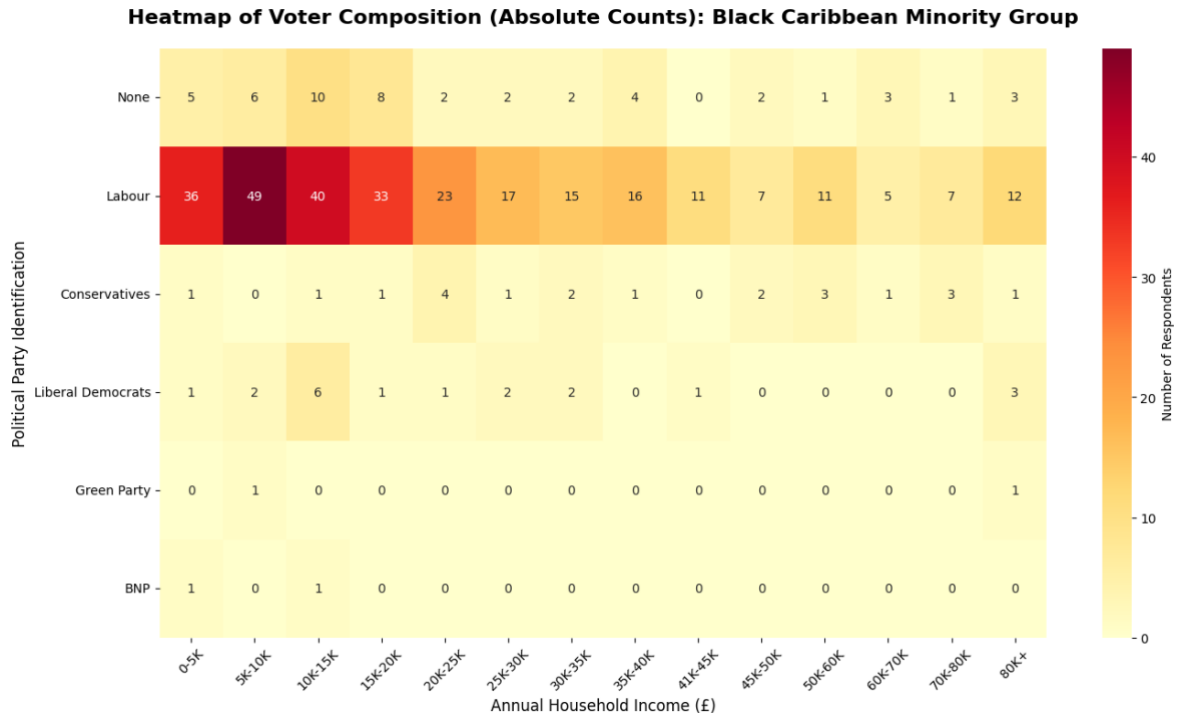
- **Bangladesh Population:**
- chi_square statistic: 35.380
- p_value:0.91189
- dof:48

[illegible]

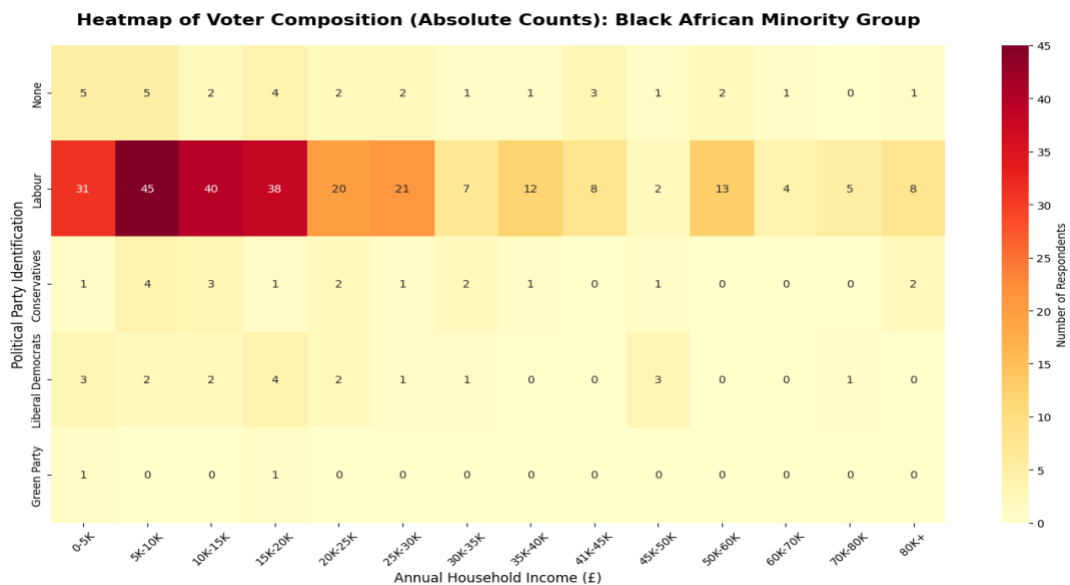
- **Pakistan Population:**
- chi_square statistic: 55.848
- p_value:0.78363
- dof:65

[illegible]

- **Black Caribbean Population:**
- chi_square statistic: 72.240
- p_value:0.25117
- dof:65



- **Black African Population:**
- chi_square statistic: 50.293
- p_value:0.54127
- dof:52



Conclutions:

Annual Household Income does **NOT** directly influence party identification among ethnic minorities ($p > 0.05$ in all income-party tests). However, the analysis show

- Income **IS** significantly associated with Ethnicity ($p = 0.0000$) - income distribution varies across ethnic groups.
- Ethnicity **IS** significantly associated with Party Identification ($p = 0.0000$) - different ethnic groups identify with different parties.
- But Income itself does **NOT** predict Party Identification ($p = 0.44/0.89$) knowing someone's income does not help predict their party choice.
- The results across all disaggregated categories are remarkably consistent. In every single case, the p-value is significantly higher than the standard critical threshold of 0.05. Particularly striking are the results for the Bangladeshi ($p=0.91$) and Pakistani ($p=0.78$) groups, which show almost total statistical independence between wealth and party choice. Even the Black Caribbean group, which represents the lowest p-value in this set (0.25), remains well within the territory of non-significance.

This shows that ethnicity is the key driver of party identification among ethnic minorities in the UK. The high p-values observed across all specific ethnic groups are entirely coherent with the final aggregate result of the study. This uniformity confirms that the lack of a relationship between income and political orientation is not a statistical anomaly of the total population, but rather a deeply rooted characteristic shared across various minority demographics

The relationship between income and ethnicity exists, but income does not translate into party preferences independently.

	Test	Chi ²	p-value	dof	Significant?
0	Ethnicity vs Closer to Party	19.450	0.07824	12	No
1	Income vs Ethnicity	194.163	0.00000	60	Yes
2	Income vs Closer to Party	38.955	0.72478	45	No
3	Income vs Party Identification	182.160	0.44097	180	No
4	Ethnicity vs Party Identification	203.141	0.00000	48	Yes
5	Income Clean vs Ethnicity	130.093	0.00000	52	Yes
6	Income Clean vs Closer to Party	27.983	0.90518	39	No
7	Income Clean vs Party Identification	134.265	0.89518	156	No
8	Voter Composition Indian Minority Group	80.379	0.40444	78	No
9	Voter Composition Bangladeshi Minority Group	35.380	0.91189	48	No
10	Voter Composition Pakistani Minority Group	55.848	0.78363	65	No
11	Voter Composition Black Caribbean Minority Group	72.240	0.25117	65	No
12	Voter Composition Black African Minority Group	50.293	0.54127	52	No

7. List all assumptions and how you addressed them.

- All variables used in the Chi-Square tests are categorical.
- When the Chi-Square value for Income vs Ethnicity was obtained, it showed a highly significant relationship. However, it was necessary to determine whether this relationship could be affected by respondents in categories -2 (Refused) and -1 (Don't Know), which are not valid income categories and contained a large portion of respondents. After running the analysis without these values, the relationship remained significant, confirming it is real and that the presence of these two groups was only inflating the result. Income distribution varies significantly across ethnic groups.
- A sufficient sample size is required, generally a minimum of $n > 50$. The dataset contains 1,732 which exceeds the minimum requirement.
- A concern arose about whether each respondent appears only once in the table or if there are duplicate IDs that could be affecting the analysis. To address this, the number of unique IDs was compared against the total number of rows, which showed that there are no duplicate IDs and each person appears only once.

8. List the null and alternative hypothesis.

	Test	H0 (Null Hypothesis)	H1 (Alternative Hypothesis)	p-value	Result
0	Ethnicity vs Closer to Party	Ethnicity and Closer to Party are independent.	Ethnicity and Closer to Party are NOT independent...	0.07824	Fail to reject H0 - No evidence of a relationship
1	Income vs Ethnicity	Income and Ethnicity are independent.	Income and Ethnicity are NOT independent.	0.00000	Reject H0 - Significant relationship exists
2	Income vs Closer to Party	Income and Closer to Party are independent.	Income and Closer to Party are NOT independent.	0.72478	Fail to reject H0 - No evidence of a relationship
3	Income vs Party Identification	Income and Party Identification are independent.	Income and Party Identification are NOT indepe...	0.44097	Fail to reject H0 - No evidence of a relationship
4	Ethnicity vs Party Identification	Ethnicity and Party Identification are indepen...	Ethnicity and Party Identification are NOT ind...	0.00000	Reject H0 - Significant relationship exists

- H0: the two variables are independent (no relationship)

- H1: the two variables are NOT independent (there is a relationship)

The decision rule is based on the p-value with a significance level of 0.05:

- If p-value < 0.05 : Reject H0 → there is a significant relationship
- If p-value ≥ 0.05 : Fail to reject H0 → no evidence of relationship

Key findings:

- Income does NOT influence whether minorities feel close to a party ($p = 0.72$) or which party they identify with ($p = 0.44$).

- Ethnicity DOES significantly influence party identification ($p = 0.00000$), meaning different ethnic groups tend to identify with different political parties.
- Income and Ethnicity ARE significantly related ($p = 0.00000$), meaning income distribution varies across ethnic groups.

In conclusion, Annual Household Income does not directly influence party identification among ethnic minorities. Ethnicity is the key factor that determines party identification.

9. Provide the results of your analysis and answer the research question.

Annual Household Income does NOT directly influence party identification among ethnic minorities. This is confirmed by p-values well above 0.05 in all income-party tests ($p = 0.44$ and $p = 0.89$), meaning we fail to reject the null hypothesis of independence between income and party identification.

However, the analysis reveals an important pattern

- Income IS significantly associated with Ethnicity ($p = 0.00000$): income distribution varies across ethnic groups.
- Ethnicity IS significantly associated with Party Identification ($p = 0.00000$): different ethnic groups tend to identify with different political parties.
- Income alone does NOT predict Party Identification ($p = 0.44/0.89$): knowing someone's income level does not help predict which party they will identify with. These results were consistent even after removing invalid response categories (-2 Refused and -1 Don't Know), confirming that the findings are robust and not inflated by non-substantive responses.

While economic differences exist between ethnic groups, these differences do not translate into different party preferences.

10. Try to explain the impact of the result within the context of the data.

The British Election Study Ethnic Minority Survey 2010 collected data from 2,787 respondents across five ethnic groups: Indian, Pakistani, Bangladeshi, Black Caribbean, and Black African in the UK.

For a long time, annual household income has been considered a factor that directly influences political decisions. However, the analysis demonstrates the opposite: income does not determine how ethnic minorities align politically. Regardless of whether a person earns £0-5,000 or £80,001+, their income level does not predict which party they identify with ($p = 0.44$) or whether they feel close to any party at all ($p = 0.72$).

In contrast, ethnicity plays a significant role. Different ethnic groups show distinct patterns of party identification ($p = 0.00000$). This suggests that cultural background, community ties, and shared experiences within each ethnic group are stronger drivers of political identity than economic status.

An interesting finding is that income distribution does vary significantly across ethnic groups ($p = 0.00000$). For example, the heatmap showed that Black African respondents are concentrated in lower income bands, while Indian respondents show greater presence in higher income bands. Yet despite these economic differences between groups, income alone does not translate into different party preferences.

This has practical implications: political parties seeking to attract ethnic minority voters should focus on culturally relevant messaging tailored to each ethnic group rather than strategies based on economic segmentation, as the political identity of these communities is shaped more by ethnicity than by income level.